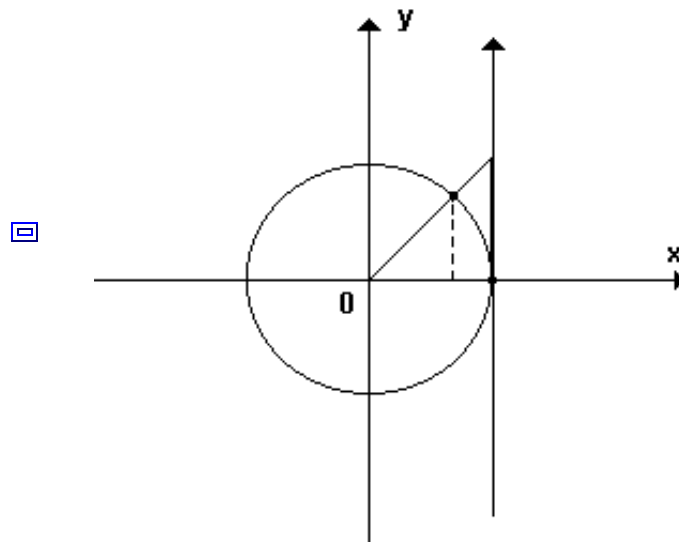


1. Limite trigonométrico fundamental

$$\lim_{x \rightarrow 0} \frac{\operatorname{sen} x}{x} = 1$$



Para $x \rightarrow 0^+$, temos $\operatorname{sen} x < x < \operatorname{tg} x$. Dividindo a dupla desigualdade por $\operatorname{sen} x > 0$, vem:

$$1 < \frac{x}{\operatorname{sen} x} < \frac{\operatorname{tg} x}{\operatorname{sen} x} \Rightarrow 1 < \frac{x}{\operatorname{sen} x} < \frac{1}{\cos x}$$

Invertendo, temos:

$$1 > \frac{\operatorname{sen} x}{x} > \cos x$$

Mas:

$$\bullet \quad \lim_{x \rightarrow 0} 1 = \lim_{x \rightarrow 0} \cos x = 1$$

$g(x) < f(x) < h(x)$ são funções contínuas e se

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = b$$

então

$$\lim_{x \rightarrow a} f(x) = b$$

Logo,

$$\lim_{x \rightarrow 0} \frac{\operatorname{sen} x}{x} = 1$$

Tabela

x	$f(x) = \frac{\text{sen}(x)}{x}$
$\pm 0,1$	0.9983341664683..
$\pm 0,01$	0.9999833334167..
$\pm 0,001$	0.9999998333333..
$\pm 0,0001$	0.9999999983333..
$\pm 0,00001$	0.9999999999833..
$\pm 10^{-10}$	0.9999999999999..
$\vdots$	$\vdots$
$x \rightarrow 0$	$f(x) \rightarrow 1$

2.

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$$

$$\text{sen}^2 x + \cos^2 x = 1, \quad \cos^2 x - 1 = -\text{sen}^2 x$$

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = \lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x + 1)}{x(\cos x + 1)} =$$

$$= \lim_{x \rightarrow 0} \frac{\cos^2 x - 1}{x(\cos x + 1)} = \lim_{x \rightarrow 0} \frac{-\text{sen}^2 x}{x(\cos x + 1)} =$$

$$= \lim_{x \rightarrow 0} \frac{\text{sen} x (-\text{sen} x)}{x(\cos x + 1)} = \lim_{x \rightarrow 0} \frac{\text{sen} x}{x} \frac{(-\text{sen} x)}{(\cos x + 1)} = 1 \cdot 0 = 0$$

$$f(x) = \text{sen} x \quad ; \quad f'(x) = \cos x$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\text{sen}(x+h) - \text{sen} x}{h} = \lim_{h \rightarrow 0} \frac{\text{sen} x \cosh + \text{sen} h \cos x - \text{sen} x}{h} = \lim_{h \rightarrow 0} \frac{\text{sen} x (\cosh - 1) + \text{sen} h \cos x}{h} =$$

$$\lim_{h \rightarrow 0} \frac{\text{sen} x (\cosh - 1)}{h} + \lim_{h \rightarrow 0} \frac{\text{sen} h}{h} \cos x = 0 + 1 \cdot \cos x = \cos x$$

$$f(x) = \text{sen}(x) \text{ então } f'(x) = \cos x$$

De maneira análoga podemos mostrar que:

$$\text{Se } f(x) = \cos(x) \text{ então } f'(x) = -\text{sen} x \quad (\text{exercício})$$

3.Limite Exponencial Fundamental

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \quad \text{ou (se colocarmos } n = 1/x) \quad e = \lim_{x \rightarrow 0} (1 + x)^{1/x}$$

x	$f(x) = \left(1 + \frac{1}{x}\right)^x$
1	2
10	2,59374246010
100	2,70481382942
1000	2,71692393224
10000	2,71814592682
100000	2,71826823720
1000000	2,71828046916
10000000	2,71828169398
100000000	2,71828178640
1000000000	2,71828203081

4. $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$ com $(0 < a \neq 1$

Façamos $t = a^x - 1$ ou seja $a^x = t + 1$. Aplicando o logaritmo neperiano em ambos os lados ficamos com $\ln a^x = \ln(t + 1)$ e assim

$$x \ln a = \ln(t + 1)$$

$$x = \frac{\ln(t + 1)}{\ln a}$$

Quando $x \rightarrow 0, x \neq 0$ temos que $t \rightarrow 0, t \neq 0$ e então teremos

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \lim_{t \rightarrow 0} \frac{t}{\frac{\ln(t+1)}{\ln a}} = \ln a \cdot \lim_{t \rightarrow 0} \frac{1}{\frac{\ln(t+1)}{t}} = \ln a \cdot \lim_{t \rightarrow 0} \frac{1}{\ln(t+1)^{\frac{1}{t}}} = \ln a \cdot \frac{1}{\ln e} = \ln a$$

Assim

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$$

Derivada de $f(x) = a^x$

$$f'(x) = \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h} = \lim_{h \rightarrow 0} \frac{a^x a^h - a^x}{h} = \lim_{h \rightarrow 0} \frac{a^x (a^h - 1)}{h} = a^x \ln a$$

$$f(x) = a^x; \quad f'(x) = a^x \ln a$$

Exemplos

1. $f(x) = 3^x; \quad f'(x) = 3^x \ln 3$
2. $f(x) = 3^x + 5x^3; \quad f'(x) = 3^x \ln 3 + 15x^2$
3. $f(x) = 3^x \cdot x^3; \quad f'(x) = (3^x \ln 3)x^3 + 3x^2 3^x$

$$(\mathbf{u} \cdot \mathbf{v})' = \mathbf{u}' \cdot \mathbf{v} + \mathbf{v}' \cdot \mathbf{u}$$

$$4. f(x) = \frac{3^x}{x^3}; \quad f'(x) = \frac{(3^x \ln 3)x^3 - 3x^2 3^x}{x^6} \quad (\mathbf{u} / \mathbf{v})' = (\mathbf{u}' \mathbf{v} - \mathbf{v}' \mathbf{u}) / \mathbf{v}^2$$